Field Engineering Tech Challenge
Candidate Brief

Candidate Name: Adrian Manchado

Demo Date: TBD

Context

The Tech Challenge is designed to lead you through the full process of creating a working deployment using the Roboflow platform. This gives you the opportunity to explore what's possible with the platform and, in turn, display what you're able to build in a short span of time using your skills and experience. This is meant to replicate a proof event (a basic PoC) that Roboflow will often deliver during a commercial engagement.

Tech Challenge

The Tech Challenge takes you from the first mile to the last mile of a computer vision deployment, culminating in a demo of your end-to-end solution using the Roboflow platform.

Use Case: Choose a use case that interests you. This will serve to highlight your industry and technical knowledge, forming the foundation of your tech challenge and demo presentation. *Pro tip:* Center the use case around a business problem that you're aiming to solve

- **Data Acquisition and Labeling:** The first mile in any AI project is data. Capture and collect relevant image/video data for your use case and upload it into Roboflow. Label the data using the various platform features. You must add your own data (local camera, public RTSP feed, etc). You are welcome to look at Roboflow Universe (or other online datasets) to augment your project dataset.
- **Model Training:** Select a project type, model size, and data augmentations to solve your problem in production.
- **Workflow Building:** Use your model(s) to build a workflow in Roboflow that helps you bring the use case to life. Your workflow must include a custom Workflow Block.
- **Deployment and Insights:** The last mile in AI projects is deployment. Deploy your workflow using Roboflow Inference and create actionable *Insights*. Common *Insights* in the field often include communicating to hardware (PLC), report building, HMIs, sending an alert, etc.

Demo

Once your project is complete, you will present and demonstrate your work in a format similar to a vendor presenting to a prospect.

- You may use additional presentation materials to support your demo as needed.
- Demo Duration: 30 minutes (hard stop)
    - The Demo of the use case should be live, not recorded
- Debrief for an additional 15-30 minutes directly after the demo for feedback and any remaining questions.

Assessment Criteria

- Technical Expertise
- Communication Skills
- Presentation Skills
- Problem-solving Skills

